



## WEEK 2, SESSION 2

# APIs you call, *webhooks that call you*.

A plain-English catalog of the services we use in the cohort, plus 50 homework ideas that combine an API with a webhook. Pick a real problem in your own life, scan for the two or three services that solve it, and build it. That is the whole assignment.

**Print this page.** File → Print. Or keep the tab open as you build. Every service below has a free tier or a test mode, so you can wire up the full loop before anyone is charged.

## SECTION 00

## How to *read this*.

Two ideas carry the whole reference.

**API** means *you call them*. Your code hits their URL with an API key, they run some work and answer back. Resend, OpenAI, Cloudinary.

**Webhook** means *they call you*. Something happens on their side, they POST to a URL you own. Stripe payments, Tally forms, Resend delivery events.

Most real integrations are both at once. You call Stripe to charge the card, Stripe calls your webhook when the payment actually settles.

**DIFFICULTY****Easy**

about 30 min.

**Medium**

one to two hours.

**Hard**three plus hours and  
real gotchas.**PRICE**

Free tier shown first, then  
the monthly paid plan. Free  
tiers are plenty for learning  
and for most small projects.

**TYPE****API****WEBHOOK**

**API + WEBHOOK**. Which half  
of the loop the service plays.

**SECTION 01**

# Communication.

Email, SMS, voice, and chat notifications. The tools you reach for when your app needs to reach a human.

| SERVICE         | TYPE          | DIFFICULTY | PRICE                         | SETUP  |
|-----------------|---------------|------------|-------------------------------|--------|
| Resend          | API           | Easy       | Free 3k/mo, \$20 for 50k/mo   | 10 min |
| SendGrid        | API           | Medium     | Free 100/day, \$20 for 50k/mo | 30 min |
| Twilio          | API + WEBHOOK | Medium     | \$0.0079/SMS, pay as you go   | 45 min |
| Discord webhook | WEBHOOK       | Easy       | Free                          | 2 min  |

## SECTION 02

## Forms & *data collection*.

The fastest way to collect input from the outside world without writing a form component. Host the form somewhere, point its webhook at your app, done.

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Week 2 API & Webhook Reference, Pioneer Species

| SERVICE                   | TYPE          | DIFFICULTY | PRICE                       |   |
|---------------------------|---------------|------------|-----------------------------|---|
| Tally                     | WEBHOOK       | Easy       | Free, unlimited forms       | ! |
| Typeform                  | API + WEBHOOK | Medium     | Free 10/mo, \$25 for 100/mo | ! |
| Google Forms (via Zapier) | WEBHOOK       | Medium     | Zapier free 100 tasks/mo    | ! |

SECTION 03

# Payments.

Taking money online is an API call to charge the card, plus a webhook to confirm the charge actually cleared. Never trust your frontend alone to decide someone paid.

| SERVICE       | TYPE          | DIFFICULTY | PRICE                     | SETUP  | BE                                                                |
|---------------|---------------|------------|---------------------------|--------|-------------------------------------------------------------------|
| Stripe        | API + WEBHOOK | Medium     | 2.9% + \$0.30 per charge  | 1 hour | In<br>st:<br>Fu<br>m<br>w:<br>bu<br>er<br>O<br>de                 |
| Lemon Squeezy | API + WEBHOOK | Easy       | 5% + processing           | 30 min | D<br>pr<br>ar<br>M<br>of<br>ha<br>V<br>sa<br>fo                   |
| PayPal        | API + WEBHOOK | Medium     | 3.49% + \$0.49 per charge | 45 min | W<br>yc<br>au<br>ex<br>wi<br>P<br>O<br>al<br>St<br>nc<br>in<br>of |



**Use test mode first.** Stripe test keys ( `sk_test_...` ) let you run the full loop with fake cards like `4242 4242 4242 4242` before you ever show a real product to a real customer.

SECTION 04

# File storage & media.

Never store user uploads in your codebase. Hand them to a dedicated service that optimizes, serves from a CDN, and bills you for storage instead of bandwidth.

| SERVICE          | TYPE | DIFFICULTY | PRICE                       | SETUP  | BEST FOR                                                         |
|------------------|------|------------|-----------------------------|--------|------------------------------------------------------------------|
| Cloudinary       | API  | Easy       | Free 25 GB, \$99 for 125 GB | 20 min | Image uploads, automatic transformations, optimized CDN.         |
| Uploadthing      | API  | Easy       | Free 2 GB, \$10 for 10 GB   | 15 min | Drop-in upload for Next.js. Simple route for working uploads.    |
| Supabase Storage | API  | Medium     | Free 1 GB, \$25 for 100 GB  | 30 min | When you already use Supabase, compatible auth layer, dashboard. |

SECTION 05

# AI & language models.

Text in, text out. Speech in, text out. Text in, speech out. You will wire at least one of these into something you build.

| SERVICE            | TYPE          | DIFFICULTY | PRICE                           | SE |
|--------------------|---------------|------------|---------------------------------|----|
| OpenAI             | API           | Easy       | GPT-4o-mini ~\$0.10 / 1M tokens | 15 |
| Anthropic (Claude) | API           | Easy       | Haiku ~\$0.25 / 1M tokens       | 15 |
| ElevenLabs         | API           | Easy       | Free 10k chars, \$5 for 30k     | 20 |
| AssemblyAI         | API + WEBHOOK | Medium     | Free 5 hours, pay as you go     | 30 |

# Automation *& workflows.*

Glue between services without writing every line yourself. Great for quick wins, but the logic lives outside your repo, so it is easy to lose track of.

| SERVICE   | TYPE          | DIFFICULTY | PRICE                         | SETUP        |
|-----------|---------------|------------|-------------------------------|--------------|
| n8n       | API + WEBHOOK | Medium     | Self-host free, cloud \$20/mo | 1 to 2 hours |
| Zapier    | WEBHOOK       | Easy       | Free 100/mo, \$20 for 750/mo  | 10 min       |
| Pipedream | API + WEBHOOK | Medium     | Generous free tier            | 30 min       |

SECTION 07

# Scheduling *& calendar.*



| SERVICE  | TYPE          | DIFFICULTY | PRICE                         | SETUP   | BEST FOR                                                                          |
|----------|---------------|------------|-------------------------------|---------|-----------------------------------------------------------------------------------|
| Cal.com  | API + WEBHOOK | Hard       | Self-host free, cloud \$12/mo | 2 hours | Open source scheduling. Embeddable into your application. Comes with API.         |
| Calendly | API + WEBHOOK | Medium     | Starts at \$10/mo             | 1 hour  | If you want a custom calendar, Calendly is a good choice. No need for an account. |



SECTION 08

Maps *& location.*

| SERVICE     | TYPE | DIFFICULTY | PRICE                | SETUP  | BEST FOR                                                                 |
|-------------|------|------------|----------------------|--------|--------------------------------------------------------------------------|
| Google Maps | API  | Medium     | Free \$200 credit/mo | 45 min | Addresses, directions, places. Requires a billing method on file.        |
| Mapbox      | API  | Medium     | Free 50k loads/mo    | 30 min | Custom map styling. Strong alternative to Google Maps if design matters. |

## SECTION 09

# Analytics *& monitoring.*

The cheapest thing you can do for the future version of yourself is make sure your app is telling you what it is doing. Page views, product events, and production errors.

| SERVICE   | TYPE          | DIFFICULTY | PRICE             | SETUP  | BEST FOR                                                                            |
|-----------|---------------|------------|-------------------|--------|-------------------------------------------------------------------------------------|
| PostHog   | API           | Medium     | Free 1M events/mo | 30 min | Product analytics, feature flags, session replay. User-behavior tracking done well. |
| Plausible | API           | Easy       | From \$9/mo       | 10 min | Privacy-friendly pageviews. Drop in the script, done.                               |
| Sentry    | API + WEBHOOK | Medium     | Free 5k errors/mo | 20 min | Error tracking. Tells you when something blew up in production, with a stack trace. |

## SECTION 10

# Quick start *picks.*

When in doubt, start here. Each of these has the easiest setup in its category.

| I WANT TO...               | USE THIS                                | WHY                                             |
|----------------------------|-----------------------------------------|-------------------------------------------------|
| Send email notifications   | Resend <a href="#">API</a>              | Easiest setup, best docs, generous free tier.   |
| Receive form submissions   | Tally <a href="#">WEBHOOK</a>           | Free, fast, no signing secret to manage.        |
| Send Discord notifications | Discord webhook                         | Literally two minutes. Paste a URL.             |
| Accept payments            | Stripe<br><a href="#">API + WEBHOOK</a> | Industry standard. Test mode works immediately. |
| Store user uploads         | Cloudinary <a href="#">API</a>          | Auto optimization and a CDN, out of the box.    |
| Add AI to your app         | OpenAI <a href="#">API</a>              | Most documented, easiest to start.              |
| Track page views           | Plausible <a href="#">API</a>           | Privacy-friendly and dead simple.               |
| Get notified of errors     | Sentry<br><a href="#">API + WEBHOOK</a> | Catches the bug before a user emails you.       |

## SECTION 11

## Integration *debugging checklist*.

When something goes wrong, walk the list top to bottom. Do not skip ahead. The thing that is broken is almost always on the first two lines.

| PROBLEM                | CHECK THIS FIRST                                                                                      | THEN CHECK THIS                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| API call fails         | Is the API key in your environment variables? Did you redeploy after adding it?                       | Open the service dashboard and confirm the request actually arrived. Then check your Vercel logs for the exact error. |
| Webhook never arrives  | Is the webhook active in the service dashboard? Is the URL correct, including <code>https://</code> ? | Submit a test (form, payment, whatever triggers it) and watch your Vercel logs in real time.                          |
| Works once, then stops | Did you hit the free tier limit? Check usage in the service dashboard.                                | Did the API key expire? Did the webhook get auto-disabled after too many failures? Re-enable and retry.               |
| Getting the wrong data | Log the actual response body. Paste it into Claude and ask it to parse.                               | Check the service's current API docs. Shapes change. Version mismatches silently.                                     |

**Log first, guess never.** In almost every case, the fastest path from "broken" to "fixed" is adding one `console.log` of the raw response and redeploying. Stop guessing what the other service sent. Make it show you.

SECTION 12

Important *notes*.

## TOPIC

## WHAT YOU NEED TO KNOW

|                    |                                                                                                                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>API keys</b>    | Always store them in environment variables, never in code. Each service uses a different prefix ( <code>re_</code> , <code>sk_</code> , <code>pk_</code> , <code>xoxb-</code> , etc.). Use test keys first, switch to live keys only when everything works.                                             |
| <b>Webhooks</b>    | The URL has to be publicly reachable. Your Vercel deploy already gives you one. Some services (Stripe) require signature verification, others (Tally, Discord) do not. Return a <code>200</code> quickly, do your real work after, or the service will retry and you will process the same event twice. |
| <b>Free tiers</b>  | Plenty for learning and most small projects. Track your usage so you are not surprised by a bill. Quick reference: Resend 3k emails/mo, Tally unlimited forms, Stripe free in test mode, OpenAI billed per token.                                                                                       |
| <b>Rate limits</b> | Every free tier has per-second and per-minute limits. Never call an API in a loop without a delay. If you start hitting limits, the service dashboard tells you which endpoint is the offender.                                                                                                         |

**A secret in the chat is a leaked secret.** If you ever paste an API key into the Claude conversation, treat it as exposed. Rotate it in the provider dashboard, update `.env` and Vercel, and move on. Future-you will thank present-you.

## SECTION 13

## 50 personal *automation ideas*.

Pick one that solves a problem you actually have. Build it for homework. The magic of this stack is that each of these is an afternoon, not a project.

### Productivity & work

| #  | AUTOMATION IDEA                                      | USES                                   |
|----|------------------------------------------------------|----------------------------------------|
| 1  | Daily task summary emailed every morning             | Resend API + scheduled function        |
| 2  | Slack ping when a high-priority task is added        | Tally webhook → Slack API              |
| 3  | Text yourself when someone schedules a meeting       | Calendly webhook → Twilio API          |
| 4  | Auto-reply to contact forms submitted after 5pm      | Tally webhook → Resend + time check    |
| 5  | Save every form submission to a Google Sheet         | Tally webhook → Google Sheets API      |
| 6  | Daily digest of messages from a Discord server       | Discord webhook → daily cron → Resend  |
| 7  | Track billable hours from a form, email weekly total | Tally webhook → database → Resend      |
| 8  | Alert when a project file is uploaded to Dropbox     | Dropbox webhook → Discord or Slack     |
| 9  | Auto-categorize expenses from a receipt photo        | Cloudinary upload → OpenAI vision → DB |
| 10 | Weekly roundup of every Notion page edit             | Notion API → scheduled job → Resend    |

## Content & social media

| #  | AUTOMATION IDEA                                  | USES                                   |
|----|--------------------------------------------------|----------------------------------------|
| 11 | Auto-post new blog posts to X                    | RSS check → Twitter API                |
| 12 | Notify yourself when someone mentions your brand | Twitter search API → Twilio SMS        |
| 13 | Save liked Instagram posts to Airtable           | Instagram webhook → Airtable API       |
| 14 | Generate a social image every time you publish   | Webhook → Cloudinary transform → post  |
| 15 | Cross-post new YouTube videos to LinkedIn        | YouTube webhook → LinkedIn API         |
| 16 | Weekly engagement stats to your inbox            | Twitter or IG API → scheduled → Resend |
| 17 | Auto-transcribe new podcast episodes             | Upload webhook → AssemblyAI → save     |
| 18 | Discord ping for every new YouTube subscriber    | YouTube webhook → Discord webhook      |
| 19 | Highlight reel from your top-performing posts    | Social API → AI → video generator      |
| 20 | Auto-reply to common DM questions                | Webhook → AI check → reply API         |

## E-commerce & money

| #  | AUTOMATION IDEA                                   | USES                                      |
|----|---------------------------------------------------|-------------------------------------------|
| 21 | Receipt email sent the moment a payment clears    | Stripe webhook → Resend API               |
| 22 | Add a new customer to your email list on purchase | Stripe webhook → Mailchimp API            |
| 23 | Text yourself when a big sale comes in            | Stripe webhook + amount check → Twilio    |
| 24 | Auto-generate an invoice PDF and email it         | Payment webhook → PDF API → Resend        |
| 25 | Fulfillment team notified in Discord              | Stripe webhook → Discord webhook          |
| 26 | Daily revenue row appended to Google Sheets       | Stripe webhook → Google Sheets API        |
| 27 | Refund triggers a feedback form email             | Stripe refund webhook → Resend + Tally    |
| 28 | 24-hour reminder on failed payments               | Stripe webhook → delayed job → Resend     |
| 29 | Low inventory alert based on recent sales         | Payment webhook → count → threshold → SMS |
| 30 | Monthly revenue report delivered as a PDF         | Scheduled job → Stripe API → PDF → email  |

## Learning & personal growth



| #  | AUTOMATION IDEA                            | USES                                     |
|----|--------------------------------------------|------------------------------------------|
| 31 | Daily quote or tip emailed every morning   | Scheduled job → API → Resend             |
| 32 | Save articles you email yourself to Notion | Email webhook → parse → Notion API       |
| 33 | Weekly learning progress report            | Form submissions → analytics → email     |
| 34 | Spaced-repetition flashcard reminders      | Database + scheduler → SMS or email      |
| 35 | Auto-transcribe voice notes to text        | Upload webhook → AssemblyAI → save       |
| 36 | Book summary when you mark one finished    | Form submit → OpenAI API → Resend        |
| 37 | Practice reminder for language learning    | Scheduled → check last practice → remind |
| 38 | Weekly digest of tweets you liked          | Twitter like → save → weekly email       |
| 39 | Journal prompt every evening at 8pm        | Scheduled → random prompt → SMS          |
| 40 | Habit tracker with streak notifications    | Daily form → track → milestone alerts    |

## Health & lifestyle

| #  | AUTOMATION IDEA                       | USES                                      |
|----|---------------------------------------|-------------------------------------------|
| 41 | Water intake reminder every two hours | Scheduled → Twilio SMS                    |
| 42 | Log workouts by SMS, weekly summary   | Twilio webhook → parse → database → email |
| 43 | Meal photo to calorie tracking        | Upload → OpenAI vision → database         |
| 44 | Medication reminder with confirmation | Scheduled → SMS → reply webhook → log     |
| 45 | Weather-based outfit suggestions      | Weather API → check temp → SMS            |
| 46 | Sleep tracker from a bedtime form     | Form → calculate hours → weekly report    |
| 47 | Daily gratitude journal prompt        | Scheduled → Resend with reply link        |
| 48 | Celebrate when you hit 10k steps      | Fitness API → threshold → confetti email  |
| 49 | Voice-note to grocery list            | Audio upload → transcribe → parse → list  |
| 50 | Birthday reminder one week out        | Database → scheduled check → SMS          |

**Rule for picking one.** If the idea makes your own life better this week, it is the right one. If you have to squint to see the value, keep scrolling. Motivation is the only thing in short supply.